

Energy Consumption Prediction using CNN-LSTM Models: A Time Series Big Data Analysis of Electricity, Heating, Hot Water, and Water

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Abstract—With the advancement of energy management systems and smart grids, the accuracy of energy consumption prediction plays a crucial role in the efficient allocation of energy resources and cost reduction. In particular, the consumption patterns of various energy resources such as electricity, heating, hot water, and water in households can vary significantly depending on the time of day and day of the week. Effectively predicting these variations is essential for managing resources. Existing energy prediction models have primarily relied on simple time series analysis techniques, which have limitations in capturing complexity of consumption patterns and long-term dependencies, thus necessitating more sophisticated prediction models. The energy prediction management system aims to more accurately forecast hourly consumption of electricity, heating, hot water, and water using a CNN (convolutional neural networks) and LSTM (long short-term memory networks) hybrid model. The data preprocessing phase utilizes a moving average technique to detect outliers and improves model performance by adding feature interactions. The system applies a CNN-LSTM model to extract local patterns from time series data using 1D convolutional layers and to learn long-term temporal dependencies using LSTM layers. The CNN-LSTM model achieves high predictive accuracy by tuning various hyperparameters, demonstrating its potential to learn the complexity of energy consumption patterns effectively. This study proves that the CNN-LSTM hybrid model can improve the accuracy of energy consumption predictions. The potential is presented for contributing to the enhancement of smart grids and energy management systems. The model provides a new technique for the efficient management of energy resources and cost reduction and suggests its applicability to real-world energy management systems.

Keywords—Smart Metering, Correlation Analysis, CNN-LSTM, Time Series Prediction, Energy Consumption Patterns

I. INTRODUCTION

Energy consumption prediction is a critical issue for efficient energy resource management and has significant implications for environmental protection. Improving energy efficiency and reducing energy demand are widely considered as the most promising, fastest, cheapest and safest means to mitigate climate change [1]. Recent advancements in machine learning and deep learning technologies have greatly

enhanced prediction accuracy for time series data, such as energy consumption forecasting in smart homes.

Various approaches have been proposed to address time series prediction's challenges. CNN (convolutional neural networks) and LSTM (long short-term memory networks) have shown considerable promise in time series forecasting due to their strengths. CNN excels at extracting and combining spatial features into higher-order patterns, making it ideal for capturing complexity of relationships in time series data [2]. In contrast, LSTM models are powerful for learning long-term dependencies in sequence data, effectively capturing and understanding the complexity of patterns in time series [3]. The CNN-LSTM hybrid model, combining these two approaches, has demonstrated superior performance in time series forecasting, and its effectiveness has been validated in recent research [4]. Moreover, recent studies have shown that anomaly detection techniques using bi-directional LSTM can significantly improve the reliability and accuracy of energy consumption data [5].

This study uses the CNN-LSTM model to predict various household energy consumption metrics based on the data analysis, including electricity, heating, hot water, and water. Since energy consumption data exhibits different patterns on weekdays and weekends.

Firstly, during the data preprocessing stage, outlier detection is performed to improve data quality, and moving averages are used to reduce data's noise. Additionally, cumulative differencing is applied to enhance data stability.

Secondly, in the feature engineering stage, not only are existing energy consumption metrics used, but also time-related features such as hour, week, month, and weekdays and weekends distinctions, as well as feature interactions among electricity, heating, hot water, and water, are incorporated into the model inputs. These features reflect the complexity of energy consumption patterns.

Thirdly, in the model training and evaluation stage, the CNN-LSTM model performs hyperparameter tuning, and various experiments are conducted to find the optimal hyperparameter combinations.

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The primary objective is to improve the accuracy of energy consumption prediction by evaluating the CNN-LSTM model's performance through different features and interactions while considering the differences between weekday and weekend data. The results contribute to developing and enhancing energy management systems and provide practical predictive results for efficient energy resource utilization. First, the processes and methodologies are described for data preprocessing and feature engineering. Second, a detailed analysis of the CNN-LSTM model's structure and hyperparameter tuning results are provided. Third, it discusses the model's practical applicability based on the results.

II. PREPROCESSING

This study employs various data preprocessing techniques to predict electricity, heating, hot water, and water consumption. Data preprocessing is essential for enhancing model training performance by transforming raw data into a format that enables the model to learn more effectively. The preprocessing methods are categorized into four main stages: data cleaning, correction, feature engineering, and data splitting.

A. Data Cleaning

To maintain dataset consistency and provide a format compatible with the model, missing values are imputed using the average of the preceding and following 10 data points. This approach helps ensure the data quality and enhances the model's predictive performance [6]. For anomaly detection, a moving average technique is applied to identify data points that are either significantly higher than the previous value or lower than the subsequent value. The moving average is calculated over a window of three data points, with a threshold set at 0.6 times the standard deviation of these three moving averages, thereby adjusting the sensitivity of the anomaly detection. Detected anomalies are then replaced with the average of the 10 preceding and succeeding data points to minimize data distortion.

B. Data Correction

The original electricity, heating, hot water, and water dataset is aggregated as cumulative values, so cumulative differencing is performed to capture the data trend. This process ensures data stability and allows the time series model to learn effectively.

When feature ranges differ, model performance may suffer. To mitigate this, we turn to MinMax-Scaling, a normalization technique that brings all features to the [0, 1] range. This normalization process significantly impacts the model by ensuring that each feature is considered equally. Consequently, the model's learning speed and convergence are notably improved [7].

C. Feature Engineering

Household energy consumption is significantly influenced by time. To incorporate this effect into the model, time-related features are added: hour, day of the week, month, and weekends and weekdays indicator. These features help the model learn about the time-dependent patterns of energy consumption.

Single features may not adequately capture the complexity of relationships between variables. By calculating correlations between features and generating interactions, as shown in

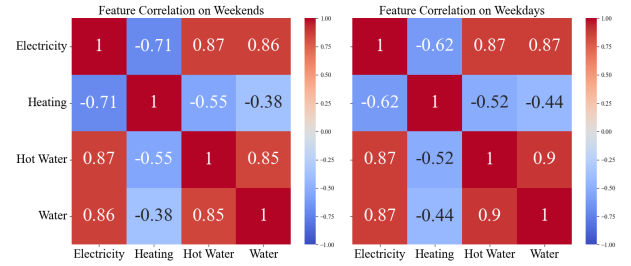


Fig. 1. Correlation between features divided by weekdays and weekends

Fig. 1, the model gains an understanding of the relationships between different variables, reflecting the complexity of energy consumption patterns.

D. Data Splitting

The data is divided into training and testing sets to train and evaluate the model. For time series data, the data is split into two halves in order to prevent data leakage and accurately evaluate the model's predictive performance.

III. MODELING

In this section, the CNN-LSTM model design is explained for time series energy consumption prediction, the hyperparameter tuning process, and the evaluation results.

A. Model Design

The CNN-LSTM model is a robust architecture that can simultaneously learn the time series data's spatial and temporal features. The CNN-LSTM model design is summarized in Table 1.

B. Hyperparameter Tuning

To optimize the model's performance, hyperparameters are tuned. This key parameter sets the dimension of the hidden states in the LSTM layers. Two values, 32 and 64, are used to compare the model's performance. Larger hidden dimensions can increase the model's representational power but may also increase the risk of overfitting [8]. The number of LSTM layers is set to 1, 2, and 3 to evaluate the model's complexity and performance. More layers can enhance the model's representational capability but may increase the training time. The number of epochs for training the model is set to 100 and 150 to ensure the model learns the data patterns sufficiently. A few epochs may result in insufficient learning, while an appropriate number of epochs is crucial to prevent overfitting [9]. Learning rates of 0.1, 0.01, and 0.001 are tested. A learning rate that is too high may prevent the model from converging, while a rate that is too low may excessively prolong training. The number of filters used in the CNN layer is set to 8 and 16 to assess the model's feature extraction capability. More filters can enhance the model's ability to recognize spatial patterns [10]. The kernel size in the CNN layer is set to 1 to effectively capture local patterns in the data, as earlier layers in the network focus on detecting simple, local features [11].

Hyperparameter tuning is performed using grid search, exploring all possible combinations to find the optimal set. As shown in Table 2, the optimal hyperparameter combinations for each target feature were identified. The tuning results optimize the model's predictive performance by balancing the risk of overfitting and improving overall generalization, thus enhancing the model's effectiveness [12].

TABLE I. CNN-LSTM STRUCTURE

LAYER	TYPE	KERNEL SIZE	STRIDE	KERNEL	INPUT SIZE	OUTPUT SIZE	ETC.
INPUT	-	-	-	-	(BATCH SIZE, 10, SEQUENCE LENGTH)	-	-
CONV1D	CONV1D	1	1	8	(BATCH SIZE, 8, SEQUENCE LENGTH)	(BATCH SIZE, 8, SEQUENCE LENGTH)	ReLU
MAXPOOL1D	MAXPOOLING	1	1	-	(BATCH SIZE, 8, SEQUENCE LENGTH)	(BATCH SIZE, 1)	KERNEL SIZE: 1
LSTM	LSTM	-	-	-	(BATCH SIZE, 64, SEQUENCE LENGTH)	(BATCH SIZE, 1)	HIDDEN DIM: 64, LAYER: 1
FC	LINEAR	-	-	-	(BATCH SIZE, 64)	(BATCH SIZE, 1)	OUTPUT DIM: 1
OUTPUT	-	-	-	-	(BATCH SIZE, 1)	-	-

C. Evaluation and Results Analysis

Each hyperparameter combination is used to train the model, and performance is evaluated using metrics such as MSE (mean squared error), MAE (mean absolute error), and R^2 (R-squared). The model trained with the optimal hyperparameters demonstrates superior predictive performance for each feature in Table 2. For the electricity, the model achieved an MSE of 0.0023, MAE of 0.0308, and R^2 of 0.7841, indicating that the model captures a substantial portion of the overall pattern of usage. For the heating, the model displayed near-perfect performance with an R^2 of 0.9972, MSE of 0.0004, and MAE of 0.0145, suggesting that it has learned the complexity of variability of the heating data with high precision. The hot water also showed low error rates, with an R^2 of 0.9656, MSE of 0.0002, and MAE of 0.0058, demonstrating that the model can accurately explain most of the variability in hot water usage. Lastly, for the water, the model recorded an R^2 of 0.9590, MSE of 0.0002, and MAE of 0.0062, indicating that it effectively captures the baseline pattern of water usage. Fig.2 visually confirms these results, where the time series plots on the left show that the predicted values closely follow the trends of the actual data. In these plots, the blue line represents the original data, the green line shows the validation data, and the red line depicts the predicted data. Additionally, the scatter plots on the right, particularly for heating and water supply, demonstrate that the predicted values are closely aligned with the actual values along the diagonal, further confirming the model's high prediction accuracy. These visualizations make it clear that hyperparameter tuning significantly improved the model's performance and that the model successfully learned the complexity of patterns for each feature.

TABLE II. RESULTS OF LEARNING WITH OPTIMAL HYPERPARAMETERS

FEAT URE	HIDD EN DIM	NUM LAYE RS	EPOC HS	LEA RNIN G RATE	NUM FILTERS	KER NEL SIZE	MSE	MAE	R ²	
ELEC TRIC	32	2	150	0.01	16	1	0.0023	0.0308	0.7841	
HEAT ING			100		8		0.0004	0.0145	0.9972	
HOT WAT ER			150				0.0002	0.0058	0.9656	
WAT ER		1	0.0002				0.0062	0.9590		

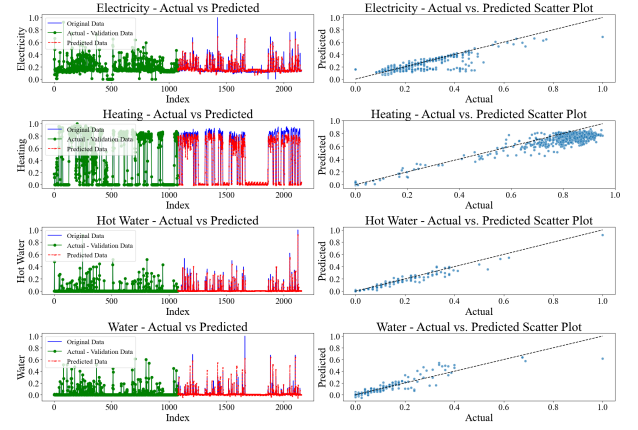


Fig. 2. Actual vs. predicted data of features time-serise plots on the left and scatter plots on the right.

IV. CONCLUSION

CNN-LSTM model is used to predict time series energy consumption data and explores how hyperparameter tuning can maximize model performance, accurately. The CNN-LSTM model has proven to be an effective tool for learning complexity of spatial and temporal patterns in time series data, resulting in improved prediction accuracy. Hyperparameter tuning optimized the model's performance, significantly enhancing the reliability and accuracy of the predictions. The CNN-LSTM model can provide a suitable solution for various time series prediction problems, effectively analyzing and forecasting the complexity of energy consumption patterns. Additionally, it highlights the impact and importance of hyperparameter adjustment on model performance, underscoring its essential role in data analysis and model development. Future research should involve comparing the CNN-LSTM model with other deep learning techniques or traditional prediction methods to assess its relative strengths, and further studies are needed to improve the model's generalization capabilities across diverse datasets. Such research will provide critical foundational insights for model development and practical applications in time series analysis, including energy consumption prediction.

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REFERENCES

- [1] S. Sorrell, "Reducing energy demand: A review of issues, challenges and approaches," *Renewable and Sustainable Energy Reviews*, vol. 47, pp. 74-82, 2015.
- [2] W. Marais, R. Holz, J. Reid, and R. Willett, "Leveraging spatial textures, through machine learning, to identify aerosols and distinct cloud types from multispectral observations," *Atmospheric Measurement Techniques*, vol. 13, pp. 5459-5480, 2020.
- [3] Y. Liu, X. Hao, B. Zhang, and Y. Zhang, "Simplified long short-term memory model for robust and fast prediction," *Pattern Recognition Letters*, vol. 136, pp. 81-86, 2020.
- [4] R. Yang, S. K. Singh, M. Tavakkoli, N. Amiri, Y. Yang, M. A. Karami, and R. Rai, "CNN-LSTM deep learning architecture for computer vision-based modal frequency detection," *Mechanical Systems and Signal Processing*, vol. 144, 106885, 2020.
- [5] H. Pan, Z. Yin, and X. Jiang, "High-Dimensional Energy Consumption Anomaly Detection: A Deep Learning-Based Method for Detecting Anomalies," *Energies*, vol. 15, no. 17, pp. 6139, 2022.

- [6] A. Karrar, "The Effect of Using Data Pre-Processing by Imputations in Handling Missing Values," *Indonesian Journal of Electrical Engineering and Informatics (IJEI)*, vol. 10, no. 2, pp. 375-384, 2022.
- [7] X. Pei, Y. H. Zhao, L. Chen, Q. Guo, Z. Duan, Y. Pan, and H. Hou, "Robustness of machine learning to color, size change, normalization, and image enhancement on micrograph datasets with large sample differences," *Materials & Design*, vol. 232, 112086, 2023.
- [8] M. Uzair and N. Jamil, "Effects of Hidden Layers on the Efficiency of Neural Networks," Department of Computer Science, National University of Computer and Emerging Sciences, Islamabad, Pakistan, 2020.
- [9] C. Xu, P. Coen-Pirani, and X. Jiang, "Empirical Study of Overfitting in Deep Learning for Predicting Breast Cancer Metastasis," *Cancers*, vol. 15, no. 7, pp. 1969, 2023.
- [10] E. Mohamed, K. Sirlantzis, and G. Howells, "A review of visualisation-as-explanation techniques for convolutional neural networks and their evaluation," *Displays*, vol. 73, 102239, 2022.
- [11] S. Kiranyaz, O. Avci, O. Abdeljaber, T. Ince, M. Gabbouj, and D. J. Inman, "1D convolutional neural networks and applications: A survey," *Mechanical Systems and Signal Processing*, vol. 151, 107398, 2021.
- [12] O. A. Montesinos López, A. Montesinos López, and J. Crossa, "Overfitting, Model Tuning, and Evaluation of Prediction Performance," in *Multivariate Statistical Machine Learning Methods for Genomic Prediction*, Springer, Cham, 2022.